

WASP-36b

R-1036

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_171124 · created 2026-07-30T07:13:36+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458081.8896 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.048
Transit depth (Rp/Rs)^2	2.78 +/- 1.59 [%]
Orbital Inclination (inc)	87.2 +/- 2.82
Airmass coefficient 1 (a1)	1.35 +/- 0.83
Airmass coefficient 2 (a2)	-0.21 +/- 1.24
Scatter in the residuals of the lightcurve fit is	61.93 %
Best Comparison Star	None
Optimal Aperture	0.99
Optimal Annulus	5.75
Transit Duration (day)	0.0309 +/- 0.0057

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.167 ± 0.048 vs 0.1368 (deviation 0.58σ)
Duration fit vs published	0.0309 d vs 0.0758 d (deviation 7.28σ)
Mid-time O–C	-5.3 min
Depth SNR	3.2
Residual scatter	61.93 %

Light curve

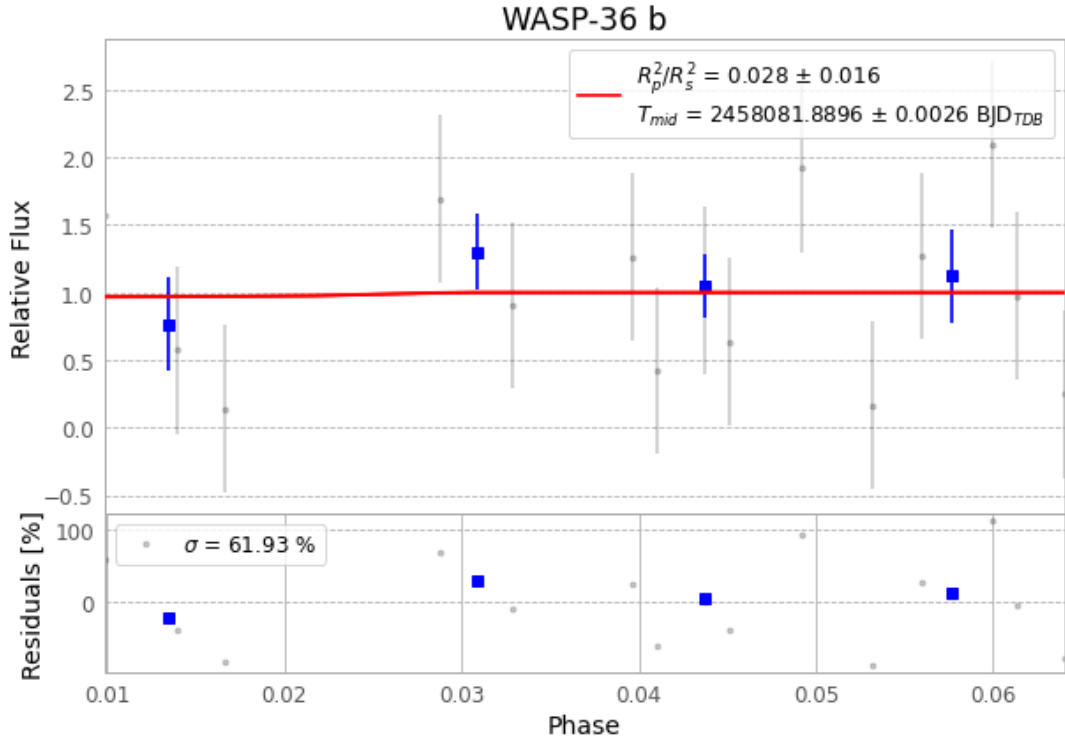


Figure 1 — FinalLightCurve_WASP-36 b_2017-11-24.png (SHA-256 52086986b42e45a6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [440, 407], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7aded0a12ba30ab7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 77e40fd

Data provenance

42 FITS frames (28 MB), WASP-36171124093312.FITS → WASP-36171124113912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-171124.log (8b3c6ea3074f...), FinalLightCurve_WASP-36 b_2017-11-24.png (52086986b42e...), FinalLightCurve_WASP-36 b_2017-11-24.csv (87489b16543e...)

Compute resources

Wall time 80.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 07:13Z.